

Mathematical Derivation of Function Value Sensitivity under Input Mean Scaling

Derivation Based on Differentiability and Chain Rule

February 12, 2026

Contents

1	Introduction	2
2	Definitions and Setup	2
3	Derivation of Sensitivity	2
4	Interpretation	3
5	Example	3
6	Conclusion	3

1 Introduction

Consider a function $f(x)$ where the input random variable x has mean d_1 . By scaling the input by factor $r = d_2/d_1$, a new function $g(x)$ is created with input mean d_2 . The goal is to express the change in function value at a specific point x_0 from $f(x_0)$ to $g(x_0)$ (sensitivity, particularly the negative sensitivity) using the derivative $f'(x_0)$, when the mean changes from d_1 to d_2 .

Assume f is differentiable at x_0 . The sensitivity is derived using the chain rule and limits for small changes in the mean.

2 Definitions and Setup

The original function is $f(x)$ with $\mathbb{E}[x] = d_1$.

After scaling the input by $r = d_2/d_1$, the new input is $y = rx$, so $\mathbb{E}[y] = rd_1 = d_2$.

The scaled function is defined as $g(y) = f(y/r) = f\left(\frac{d_1}{d_2}y\right)$, ensuring the function adjusts to the new input scale.

At the specific point x_0 in the new input space, $g(x_0) = f\left(\frac{d_1}{d_2}x_0\right)$.

The change in value is $\Delta = g(x_0) - f(x_0) = f\left(\frac{d_1}{d_2}x_0\right) - f(x_0)$.

The sensitivity (rate of change) with respect to the mean change $\Delta d = d_2 - d_1$ is approximated as $\frac{\Delta}{\Delta d}$.

3 Derivation of Sensitivity

To find the sensitivity, consider the function $g(d_2) = f\left(\frac{d_1}{d_2}x_0\right)$ and compute its derivative with respect to d_2 at $d_2 = d_1$:

$$\left. \frac{\partial g(x_0; d_2)}{\partial d_2} \right|_{d_2=d_1} = \lim_{d_2 \rightarrow d_1} \frac{g(x_0; d_2) - g(x_0; d_1)}{d_2 - d_1} = \lim_{d_2 \rightarrow d_1} \frac{f\left(\frac{d_1}{d_2}x_0\right) - f(x_0)}{d_2 - d_1}.$$

Let $u(d_2) = \frac{d_1 x_0}{d_2}$, so $g(d_2) = f(u(d_2))$.

The derivative of u is $u'(d_2) = -\frac{d_1 x_0}{d_2^2}$.

By the chain rule,

$$g'(d_2) = f'(u(d_2)) \cdot u'(d_2) = f'\left(\frac{d_1 x_0}{d_2}\right) \cdot \left(-\frac{d_1 x_0}{d_2^2}\right).$$

Evaluate at $d_2 = d_1$:

$$u(d_1) = x_0, \quad g'(d_1) = f'(x_0) \cdot \left(-\frac{d_1 x_0}{d_1^2}\right) = f'(x_0) \cdot \left(-\frac{x_0}{d_1}\right) = -\frac{x_0}{d_1} f'(x_0).$$

Thus, the sensitivity is

$$\frac{\Delta g(x_0)}{\Delta d} \approx -\frac{x_0}{d_1} f'(x_0).$$

The change in value is

$$\Delta \approx -\frac{x_0}{d_1} f'(x_0) \cdot (d_2 - d_1).$$

The negative sign indicates that for an increasing mean ($d_2 > d_1$) and positive $f'(x_0)$, the function value decreases at x_0 (negative sensitivity).

4 Interpretation

The sensitivity $-\frac{x_0}{d_1}f'(x_0)$ quantifies how the function value at x_0 changes with the input mean scaling. The term $\frac{x_0}{d_1}$ represents the relative position of x_0 to the original mean, and $f'(x_0)$ is the local slope.

For small Δd , this is a first-order Taylor approximation. Higher-order terms may be needed for larger changes.

5 Example

Consider $f(x) = x^2$ ($f'(x) = 2x$), $d_1 = 1$, $d_2 = 1.1$ ($r = 1.1$), $x_0 = 1$.

Then $g(x) = f(x/1.1) = (x/1.1)^2$.

$g(1) = (1/1.1)^2 \approx 0.8264$, $f(1) = 1$, $\Delta \approx -0.1736$.

Approximation: $-\frac{1}{1} \cdot 2 \cdot 1 \cdot 0.1 = -0.2$ (close, small error for $\Delta d = 0.1$).

6 Conclusion

The change in function value due to mean scaling from d_1 to d_2 at x_0 is approximated by $-\frac{x_0}{d_1}f'(x_0)(d_2 - d_1)$, derived via the chain rule. This expresses the negative sensitivity using $f'(x_0)$.